The Journal of Open Source Software

pyopmspe11: A Python framework using OPM Flow for the SPE11 benchmark project

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DOI: 10.xxxxxx/draft

Software

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Editor: Rachel Wegener

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- @MatthewFlamm
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Submitted: 28 June 2024
Published: unpublished

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Summary

The imperative to achieve climate change goals and the increasing worldwide demand for energy have made carbon capture and storage (CCS) technology more relevant today. Since utilizing computational models is essential for planning large-scale CCS projects, it is crucial to benchmark simulation tools to enhance confidence in their results. Inspired by a recent validation study for laboratory-scale CO_2 storage (Flemisch et al., 2024), a new comparative solution project (CSP) was launched to simulate both lab- and field-scale CO_2 storage (Nordbotten et al., 2024). As part of the Early Access team, the authors have developed and made open the `pyopmspe11` tool which facilitates reproducible solutions to the SPE11 benchmark. This tool serves as a common starting point for developing and testing new simulation technology. Due to its user-friendly functionality (e.g., generation of different type of grids at different grid resolutions, flexibility to choose different rock and fluid properties, flexibility to define well/source locations and schedule for operations), then it is expected that its impact will extend far beyond the initial benchmark study (e.g., studies focusing on grid refinement, upscaling/coarsening approaches, numerical solvers, optimization/history matching techniques).

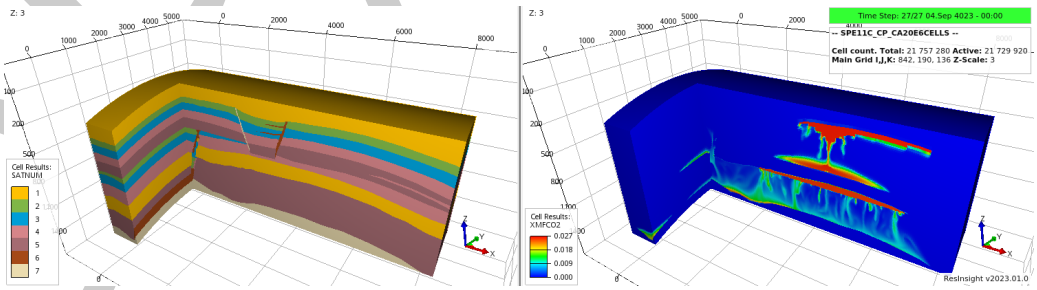

Figure 1: Generated model by the configuration file `spe11c_cp_ca20e6cells.txt` in the examples folder.

Statement of need

Geological carbon storage (GCS) applications benefit from both commercial and open-source simulators. However, using numerical simulators often involves data preprocessing and post-processing, which can be challenging for everyone even if you know what you are doing. Additionally, setting up and running simulations requires computational expertise. To bridge this gap, developers can simplify the setup of numerical studies by using user-friendly approaches, such as configuration files. This not only ensures reproducibility of results but also facilitates flexible testing of different simulator parameters and allows for easy extension to further studies. Notable works are available in JOSS for pre-/postprocessing of simulation data, e.g., Beucher et al. (2019), Sullivan & Kaszynski (2019), Fraters et al. (2024), Kaus

et al. (2024). Examples of tools which application is also in GCS and relies on the OPM Flow simulator are `pyopmnearwell` (Landa-Marbán & pschultzendorff, 2023) and `expreccs` (Landa-Marbán, 2024). The former focuses on near well dynamics, while the latter on seamless, dynamic, and non-invasive exchange of pressure-related information between local and regional scales.

Based on the acquired knowledge by developing the aforementioned tools, as well as from using/contributing to other open-source projects, then we have developed and made open the `pyopmspe11` tool which facilitates reproducible solutions to the SPE11 benchmark, which focus on GCS at different scales (Nordbotten et al., 2024). A previous benchmark study for GCS can be found in Class et al. (2009). One key difference of the SPE11 benchmark from the previous one is that no grids were given in the description, i.e., one of the main task for the participants was to create suitable grids (e.g., structured grids such as Cartesian or unstructured grids such as corner-point grids). The participants were encouraged to share data (e.g., input decks, code, submitted results), with the opportunity to store the data for open access. This is where developing tools that made all steps reproducible (i.e., preprocessing and postprocessing) become handy, and for this benchmark study, one available tool is `pyopmspe11`.

`pyopmspe11` is a simplified and flexible Python tool to execute the three cases in the SPE Comparative Solution Project using configuration files. `pyopmspe11` relies on the OPM Flow numerical simulator (Rasmussen et al., 2021), where the implementation of the CO_2 model can be found in Sandve et al. (2021). The primary contribution of `pyopmspe11` lies in its data preprocessing and postprocessing capabilities. It offers flexibility in generating various types of grids, including Cartesian, tensor, and corner-point grids. These grids adhere to the standard industry format (i.e., Eclipse grid format), making them compatible not only with OPM Flow but also with other simulators. Here, we mention two existing widely-use visualization/postprocessing software for OPM Flow: ParaView and ResInsight. While these tools are very useful, to the authors knowledge, there are no available built-in options via the GUIs of these tools to handle all necessary postprocessing to generate all data reporting as required in the SPE11 benchmark study; instead this could be achieved in these tools by code development using the Python interface. The benefit of having a specilized tool like pyopmspe11, which integrates all stages from preprocessing to simulation and postprocessing, is that it simplifies usage.

`pyopmspe11` supports varying resolutions, having been tested to generate up to 160 million cells. In the context of data postprocessing, `pyopmspe11` not only generates the necessary reporting data as specified by the benchmark, but it also produces .png figures for rapid inspection of individual simulations and for making comparisons between different runs (e.g., to assess sensitivities). The postprocessing methods efficiently interpolates quantities over time and map non-overlapping cell values (both intensive and extensive quantities) between the simulation grid and the reporting grid. The Python package Scipy (Virtanen et al., 2020), specifically the interp1d Class, is used for the time interpolation. The Python package Shapely (Gillies et al., 2024), specifically the Polygon Class, is the base for the developed methods in `pyopmspe11` to handle the mapping from the simulation grid to the reporting grid.

Outlook

`pyopmspe11` is designed for use by researchers, engineers, and students. During the preliminary intercomparison workshops for the benchmark study, the authors received positive feedback about the framework. Some participant groups have utilized `pyopmspe11`, particularly for grid generation. Additionally, the authors have been contacted to provide specific support on how to use the tool for setting up further studies, such as optimizing injection strategies. Looking ahead, the plan for `pyopmspe11`'s future development includes extending its functionality to support the generation of input decks to run simulations of physical models available in OPM Flow in addition to CO_2 storage (e.g., hydrogen storage, salt precipitation, biofilm effects).

Acknowledgements

The authors acknowledge funding from the Center for Sustainable Subsurface Resources (CSSR), grant nr. 331841, supported by the Research Council of Norway, research partners NORCE Norwegian Research Centre and the University of Bergen, and user partners Equinor ASA, Harbour Energy, Sumitomo Corporation, Earth Science Analytics, GCE Ocean Technology, and SLB Scandinavia. The authors also acknowledge funding from the HPC Simulation Software for the Gigatonne Storage Challenge project, grant nr. 622059, supported by Equinor ASA and CLIMIT DEMO/Gassnova.

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